Stage 2 Report

1. Exploration Goals

This stage involves exploring, analyzing, and cleaning the football player dataset to identify missing values, redundancies, and inconsistencies and improve the overall quality of the dataset. The refined data will support predicting the **Football Player Future Potential Score (0–100)** and classifying players as **Outperforming, Average, or Underperforming**, completing essential preparation for preprocessing and model development.

2. Dataset Overview

The merged football dataset originally contained **50,149 rows** and **73 columns** collected from multiple Transfermarkt data sources. After analyzing column relevance, redundancy, and missing values, unnecessary columns were removed during the cleaning process.

The final cleaned dataset contains:

- **26 Numerical columns**
- **18 Categorical/Text columns**
- **Total Columns:** 44 (including 2 target columns: 1 numerical and 1 categorical)
- **Total Rows:** 50,149

The dataset includes player profile information, match performance statistics, market value details, transfer history, club information, competition details, and the target variables required for predicting a player's future potential.

3. Data Quality Assessment (MIDIO)

Missing Values

Several columns contained missing values. Some columns such as **total_market_value** contained 100% missing values and were removed from the dataset because they provided no useful information. Certain columns related to transfer history contained missing values because not every player had been transferred. Such missing values were considered logically valid.

Duplicates

Duplicate records were checked in the merged dataset, and no duplicate rows were found. Therefore, no duplicate records were removed during the cleaning process.

Inconsistencies

The dataset was examined for inconsistent values, duplicate metadata created during dataset merging, and unnecessary identifier columns. Redundant columns and metadata fields were removed to improve data quality.

Outliers

Boxplots were used to identify outliers, while histograms were used to examine the distribution of important numerical attributes such as market value, goals, assists, minutes played, transfer fees, and player height. Since these values represent real-world player performance (for example, elite players with exceptionally high market values), they were retained rather than removed.

4. Cleaning Decisions and Justification

- Identifier columns such as **player_id**, **club_id**, and other ID fields were removed because they do not contribute to prediction.
- Duplicate records were checked and no duplicate rows were present in the final dataset.
- URL, image, filename, and metadata columns were removed because they have no analytical value.
- The **total_market_value** column was removed because it contained 100% missing values.
- Important football-related columns such as player profile, market value, performance statistics, transfer history, club information, competition details, and target variables were retained because they are relevant for predicting player future potential.

These cleaning decisions improved the quality of the dataset while preserving information required for machine learning.

5. Exploratory Data Analysis

The following visualizations were performed to understand the dataset:

- Histograms of important numerical features such as market value, goals, assists, and minutes played.
- Boxplots to identify outliers in player performance and market value.
- Bar charts showing the distribution of categorical variables such as playing position and preferred foot.
- Distribution plots for **Player Future Potential Score** and **Player Future Potential Rating**.
- Correlation heatmap to examine relationships among numerical features.
- Scatter plots to study the relationship between important performance metrics and the target variable.

These visualizations helped identify trends, feature distributions, relationships, and potential anomalies within the dataset.

6. Stakeholder Impact of Clean Data

Football Clubs

A clean dataset helps football clubs evaluate player potential more accurately before making transfer and recruitment decisions.

Football Scouts

Reliable player statistics allow scouts to identify talented players with strong future potential using objective data.

Player Agents

Accurate performance and market value information helps player agents assess future player potential during contract negotiations and transfer discussions.

7. Conclusion

The data exploration and cleaning process improved the quality of the football player dataset by removing irrelevant columns, eliminating duplicate records, removing the completely empty column, and retaining logically valid missing values. The cleaned dataset now contains **50,149 rows** and **44 columns**, making it well-structured and suitable for feature engineering, preprocessing, and predictive model development in **Stage 3**.